

Family life in the digital home — domestic telecommunications at the end of the 20th century

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This paper describes results from the first stage of a long-term study of people's use of ICT (information and communications technology) products and services in the domestic environment. It is part of BT's ongoing commitment to understanding, modelling and meeting the needs of its residential customers. At one level the research has already generated significant knowledge that has led directly to commercial benefit; at a second it has validated a number of research methods needed to capture the data to build a rich picture of people's actual as well as reported behaviour. At a third level it has also generated significant new scientific knowledge about the use of ICT products.

*The next step is to build on these successes using a 1000 strong UK household panel to build an unprecedented and unparalleled understanding of how and why people **purchase**, **adapt** and **adapt to** ICT as UK society moves into the 21st century.*

1. Introduction

The UK residential telecommunications market is becoming increasingly competitive, and an understanding of customer loyalty is vital if an incumbent is to maintain its existing customer base, increase its market share, and retain competitive advantage throughout the digital revolution. Among other factors, customers are more likely to remain loyal to a company if the range of products and services on offer fit their needs and life-styles.

Most telecommunications companies operating in the UK, including BT, commission market research with a wide sample of the UK population about aspects of communications technologies, both on an event-driven basis (for example, to provide information for a new product launch) and on a regular basis. Traditional market research techniques are a valuable means of capturing current consumer behaviours and attitudes. However, they can be poor at providing insight into future behaviours because it is difficult for respondents to predict what they intend to do with any accuracy and there is often a difference between reported and actual behaviours [1]. In addition, there has been little comprehensive research on the actual use of information and communications technology (ICT) within the home because traditional surveys have concentrated on data capture and analysis at the gross population level rather than empirically studying real patterns of use.

The 'Family Life in the Digital Home' project has piloted and applied novel qualitative methodologies and

sophisticated predictive modelling techniques to investigate naturally occurring usage of communications and information technologies by individuals in the domestic context over significant periods of time.

The project consists of two phases:

- a recently completed year-long study that assessed a variety of data-capture techniques,
- a three-year implementation of successful techniques to address research questions arising from the literature, the first year's work, and market drivers.

2. Related work

The impact of the rapid growth of the information and communication technology industry on society is an issue that has generated significant research efforts in both the commercial and academic sectors.

Despite this, data on the domestic usage of the telephone in the UK is extremely scarce. One notable exception is Frolich et al's analysis of 315 household telephone calls [2]. However, a small number of studies such as de Sola Pool [3], Marvin [4] and Fischer [5] provide a picture of the social and commercial evolution of telephony in the United States.

Surveys such as that by the NOP Research Group take a wave-on-wave approach to investigate issues including the latest Internet user profiles, children's use of the Internet,

and how the Internet is used by businesses¹. The INTECO Corporation conducts extensive research to analyse interactive technology markets and services across the USA and Europe, including the Internet, PC, televisions, cellular telephones and electronic commerce platforms², while Georgia Tech's Internet Users survey provides a wealth of mainly US-centric information on Internet usage gathered on-line from a self-selecting sample³.

The HomeNet⁴ project, led by Professor Robert Kraut at Carnegie Mellon University, Pittsburgh, USA, is probably the only significant attempt to systematically study ICT usage over time in a manner similar to the Family Life project.

Over 100 households in the Pittsburgh area have been supplied with computer equipment and Internet accounts, and the programme is using a combination of questionnaires and electronic data collection to explore Internet usage and its effects on participants' lives. By providing the technology and Internet service for free, Kraut's team has removed economic barriers to usage. This allows research to be conducted with families who could not usually afford to access the Internet, but as a result the study of usage and uptake is made somewhat artificial [6].

The European Internet market is less well developed than in the USA, where the main point of Internet access has been the home. By mid-1998, 58% of all adult Internet users had access from home whereas less than half of all users in the major European countries had access from home. This is partly because of the low penetration of home PCs and modems in Europe — home-PC penetration in the major European countries stood at roughly half the USA level of 42% at the end of 1997. To some extent this is due to perceived relatively high local telecommunications costs in Europe compared with the USA, although it should be noted that growth in home Internet access in the USA almost came to a halt in mid-1996 once the uptake by early adopters had levelled out. It was not until flat-rate pricing was introduced that the market took off again.

The Family Life project builds on the above studies in a number of respects.

- The project aims to understand all aspects of information consumption and communication within the home that is mediated by technology, be it telephony, e-mail, on-line services, PC, television or some other product or service which emerges during the study period. Gaining an understanding of the interdependencies of different media will be a key output of the project.

- The project intends to follow the same individuals throughout the three years of the full study rather than re-sampling for each wave of the research. This will enable us to measure the take-up of new technologies in a rigorous manner and enable an understanding of factors driving their usage.
- The project is devising and applying a wide variety of different data-collection and analysis techniques to provide insight into the rationale behind observed behaviours at the micro level, rather than relying on interviews and questionnaires alone.
- The project is also using background customer information which can provide further demographic and behavioural data to provide an extremely 'rich picture' of residential communications behaviour. This 'multi-method' approach will provide the best opportunity to understand the complexities of communications behaviour, including the underlying context and rationale.

3. Methods

During the first year of the Family Life project, a number of individual household usage studies were carried out using a variety of data collection and analysis techniques. This section outlines the methods used in three of the main studies.

3.1 Pilot household study

A sample of 367 households were selected from the British Household Panel Survey (BHPS), an existing panel study administered by the Institute for Social and Economic Research at the University of Essex. The BHPS was established as a nationally random sample in 1991 and has been repeated in yearly waves.

The Family Life sample was selected with the aim of understanding ICT usage within the home, and the social patterns underlying this usage. A secondary objective of the work was to assess whether a sample of this nature would provide an appropriate 'backbone' for longitudinal panel research over the subsequent years of the Family Life project.

Two different data collection techniques were used on the sample — questionnaires and time-use diaries. 75% of eligible individuals from the sample households were interviewed, giving a total of 699 interviews. In addition, 83% of interviewed adults returned 582 completed time-use diaries. 46% of the sample were male, and 54% were female. Table 1 summarises the socio-economic categories referred to here and later in this paper.

¹ <http://www.nopres.co.uk/internet.asp>

² <http://www.intecoresearch.com/public/index.htm>

³ <http://www.gvu.edu>

⁴ <http://homenet.andrew.cmu.edu/progress/>

Table 1 Socio-economic categories.

Social grade	% of adult population	Head of household occupation
A	3	Higher managerial, administrative or professional.
B	14	Middle managerial, administrative or professional.
C1	22	Supervisory or clerical, junior managerial.
C2	31	Skilled manual workers.
D	19	Semi- and un-skilled manual workers.
E	11	State pensioners, casual or lowest grade workers, unemployed.

Table 2 describes the sample used in this stage of the research in terms of age, socio-economic status and family distributions.

Table 2 Sample profile giving percentage of each category.

Age	%	Socio-economic	%	Family	%
≤20	6	AB	16	Couples without children	36
21—30	23	C1	35	Couples with children	29
31—40	16	C2	13	Lone parent families	8
41—50	16	D	17	People over 50 living alone	16
51—60	17	E	19	People under 50 living alone	7
>60	22			Some other arrangement	4

Questionnaires

The questionnaires, comprising a 'Household' and an 'Individual' questionnaire, were administered on a face-to-face basis. The 'Household' questionnaire was administered to one adult household member. The questions covered household membership, tenure and housing costs, consumer durables, number of televisions, telephones and computers and their locations, information on add-on devices, telephone bills, and car ownership.

The 'Individual' questionnaire was administered to every household member aged 16 or above. Respondents were asked about their general attitudes towards technology, various aspects of computer usage at home and at work, telephone contact with friends and relatives, education and training, newspapers and other readership, leisure activities, division of labour within the home and standard demographic questions.

Time-use diaries

All adults aged 16 or above were asked to complete diaries to record how much time they spent over a week on over thirty pre-specified activities (covering domestic routines, work, outdoor and indoor leisure, telephone usage, and different types of computer usage). A version for children aged between 9 and 15 years was adapted from the adult diary.

3.2 Qualitative anthropology

A series of 48 in-depth semi-structured interviews were conducted with members of six households to establish how people define their communications needs in the home, the significance of communication in their lives, and their current actual usage. The study began with the concept of a communications network, telephone users and the people with whom they talk most regularly on the telephone. The main users were interviewed in depth in their own homes, as were the people they nominated as being their most common telephone contacts. Respondents were asked about their life-style, relationships with family and friends, the size of their calling circle, what they talk about, calling patterns, routines, and rules about usage. Telephone bills for the previous three months were also examined, and, in households with more than one member, participants were asked to say who made which calls. Participants were selected to include a variety of life experiences and situations in order to give as broad a perspective as possible. In addition one group was shadowed for two complete days with the researcher collecting data on all incoming and outgoing calls together with all other communications-related activities.

3.3 Home communications diary study

This study was carried out with the aim of providing an understanding of the impact of physical location on product usage. It also served as an additional trial of the diary methodology, and enabled the use of more innovative tools and techniques for understanding household communication patterns.

Sixty British households were asked to record all incoming and outgoing calls for both telephony and data communications products in a communications diary. The diaries were administered over three separate seven-day periods between February and April 1997. Participants were also asked to use a disposable camera to take photographs of their communications products, their houses and household members, and any other photographs they felt would reflect their life-styles in general ⁵.

The households were selected from an existing BT trial of a new telephony product and so were not truly

⁵ As all participants were guaranteed anonymity and confidentiality, examples of the photographs cannot be reproduced in this paper.

representative of the British population. The sample was heavily biased towards technology and communications-aware households — for example, 76% of respondents owned a PC, 75% had a mobile telephone, and 27% had two or more telephone lines. Just under a third (31%) had cable or satellite.

Overall, 55 of the 60 households completed the communications log during week one, and 43 households completed the communications diary over all three weeks. This compares very favourably with the typical response rate of around 33% for postal surveys. A total of 5,384 calls were logged in the diaries over the three weeks of the survey.

Analysis of the data from the three separate studies revealed a number of insights into communications behaviours and usage patterns. These are discussed in the following sections.

4. Results

This section describes the results from the first year of the project. The results are presented in terms of themes that became apparent as the research progressed and which will be explored further in phase two of the project.

4.1 The demographics of ICT ownership and use

Income and social class

The results confirm findings from other research which show the general pattern of ICT ownership and usage being driven by income and social class.

There is a strong correlation between social class and ownership of PCs, telephones and televisions as is clearly shown by Fig 1.

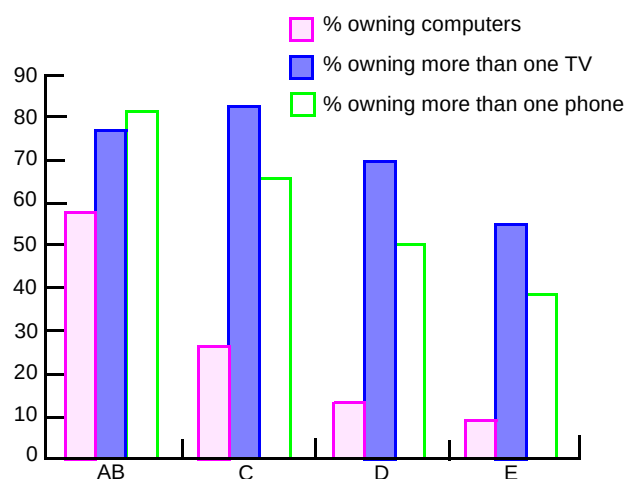


Fig 1 Relationships between social class and ICT ownership.

Usage of the telephone follows the same general pattern with usage in terms of hours on the telephone increasing as income increases. The same pattern emerges when considering bill size. PC usage does not seem to correlate

positively with social class; of those who have a PC, people in socio-economic class D use it more than Cs who use it more than ABs as is shown by Fig 2.

The survey results indicated that those with computers were more likely to keep in contact with friends by telephone, but less likely to keep in touch with relatives by telephone. In the case of contact with friends, it appears that high telephone contact and high computer usage reinforce each other. However, computer users are less likely to have regular contact with relatives, whether by telephone or face to face, than are non-computer users.

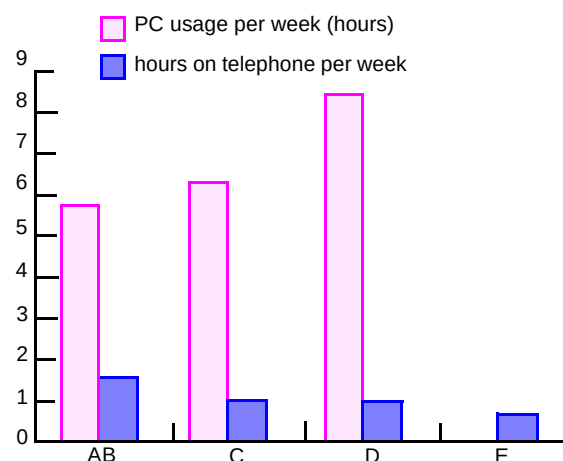


Fig 2 Relationship between social class and ICT usage.

Rather than suggesting a direct link between computer usage and sociality, it is more likely that these findings reflect differences in communications behaviour according to socio-economic status. It is known that ABs typically spend longer on the telephone than C2Ds and Es and that PC ownership is strongly related to monthly household income and social class. Argyle and Henderson [7] contend that people from the higher socio-economic groups have more friends, who are more diverse and live at greater distances, which implies that a substantial proportion of their communication will be over the telephone. In contrast, they contend that lower socio-economic groups tend to communicate with friends on a face-to-face basis rather than via the telephone. In addition, 'working class women see a lot of their kin', usually close female relatives such as mothers, aunts and sisters, on a regular daily basis which might explain higher telephone contact with relatives.

Family structure and life-stage

It appears that family structure is a strong predictor of computer ownership since 44% of couples with children owned a PC, 40% of young singles and 29% of childless couples. Only 25% of lone parents own a computer perhaps reflecting the influence of lower relative income. The presence of children appeared to make some difference to computer usage with the diary data indicating that in

computer-owning households, couples without children used computers for an average of 7.6 hours per week, compared to 8.6 hours per week for couples with children. The questionnaire data indicated that couples tended to share usage of their computer, but this was not corroborated by the diary data, which indicated that differences in computer usage patterns for men and women outside partnerships were far smaller than for couples (whether or not they had children). This illustrates an interesting gap between stated and actual behaviours, and reinforces the case for using a variety of different data collection methodologies.

Having children made very little difference to the number of telephones owned; 33% of couples without children had three or more telephones, compared to 30% of couples with children, although couples with children were more likely than those without to have a mobile telephone. It appears that whether or not people are single or with a partner affects the number of telephones owned rather than the presence of children, as single people are much more likely to have one telephone only. Perhaps surprisingly, telephone bill size did not appear to be strongly related to family situation; for example, the average telephone bill for couples with children was £88, compared to £82 for those with no children.

4.2 Usage and gender

The studies found strong evidence of apparent gender differences in usage of computers by couples (see Fig 3). In addition the diary data also showed that men with at least some recorded computer time spent an average of 9.9 hours per week using a computer at home, compared to 5.6 hours for women. Men appear to access the Internet for longer periods than women (2.4 compared to 0.2 hours per week) and spend an average of 3.4 hours per week playing computer games, compared to 1.2 hours for women.

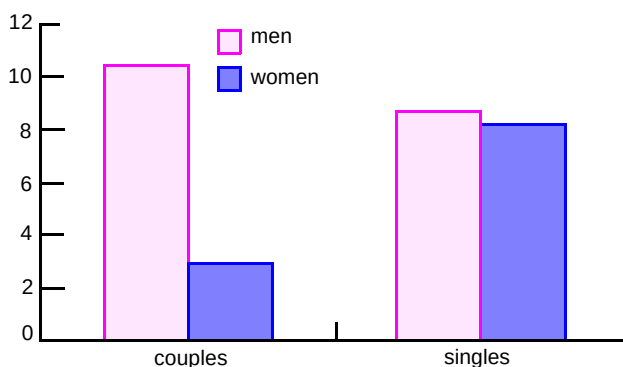


Fig 3 Mean hours using PC per week derived from time-use diaries.

The questionnaire data suggests that partners (married or not) did have a tendency to share telephone habits. For example, where a woman reported phoning relatives most days, so did 60% of their partners. However, this is not

confirmed by the diary data as Fig 4 illustrates. The diary data makes clear that women are more likely to keep in daily telephone contact with relatives, but there is a much smaller difference with regard to phoning friends on a daily basis.

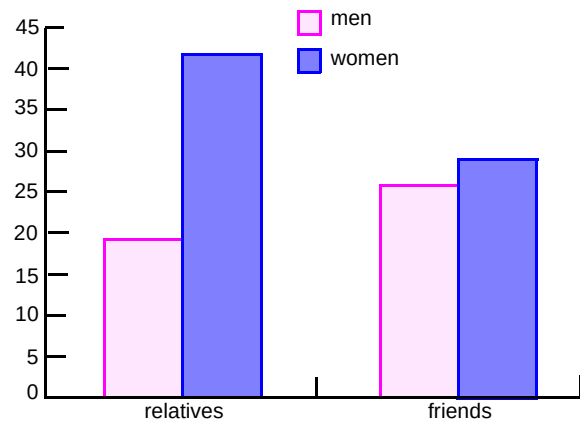


Fig 4 Percentage calling friends and relatives daily by gender (Pilot Household diary).

This finding is supported by data from Home Communications diary study, which found that adult females were involved in nearly half as many incoming and outgoing calls as adult males. Results from the qualitative study also indicated that women used the telephone more frequently.

In terms of call duration, the Pilot Household diary data indicated the gender difference was as strong for total duration as for frequency. Women spent an average of 1.1 hours per week on the telephone, compared to 0.7 hours per week for men. The qualitative study found that women tended to make longer calls, and have larger calling circles that combine to produce their higher usage rates. This may be explained in part by the fact that women tend to be the dominant influence in maintaining communication networks with family and friends. Several of the women in the study said they often prompt their partners to make calls, or often make them on their partner's behalf as shown in this comment from a female participant talking about her partner:

'If he hasn't rung his dad for a while I'll nag him for a bit and then I'll usually do it myself'.

This is consistent with Argyle and Henderson [7] who comment that men form less close attachments to friends, with less self-disclosure. Their attachment is often formed with a group of people, and they spend their time on shared activity rather than intimate conversation.

The qualitative study also found differences in terms of the actual content of men's and women's calls. Men tend to use the telephone to make arrangements, whereas women tend to use the telephone more for social communication (chatting), and the content of the call is more diverse. Men

also claim that they always know what they are going to talk about before making a call, i.e. calls are made with a specific purpose in mind and are brought to a close once that purpose has been achieved. Women are more likely to make a call 'just to chat' and, even when having a specific purpose in mind, are much more likely than men to deviate from this and discuss other unrelated matters. Although both male and female participants claimed that the telephone is one of life's essentials, there was a clear disparity between men and women's view of what the telephone is for. The men claimed that it is primarily a tool for checking and making arrangements, whereas the women believed it to be vital for the maintenance of their social relationships. For example, a male participant talking about his own and his partner's calls commented:

'She makes more personal calls, calls to friends and family, often for things in particular but she ends up just chatting. My calls tend to be connected with what's going to happen, to make arrangements. I do chat on the telephone sometimes but I prefer face-to-face.'

The Home Communications diary study also found that women used communications products as an information source, to chat, and for social and domestic arrangements, whereas men used communications products primarily as an information source and for social and domestic arrangements rather than chatting. The fact that the Pilot Household diary results suggested that many men appear to make very short calls supports these findings.

The qualitative study found that men and women appear to vary in their attendance to and dependence on verbal and non-verbal cues. Women were more likely than men to comment on such cues in the course of the conversation. Almost all the men in the study said that they preferred face-to-face communication to talking on the telephone and located some of their discomfort in using the telephone in not being able to see the reaction of the person to whom they were speaking. Two comments from male participants illustrate these points:

'Quite often if I'm talking to a woman on the telephone they're quite likely to say how they think I'm feeling. Even when I know I sound down and I'm talking to another guy he wouldn't say anything about it.'

'You can't pick up cues because you can't see their face and it's more difficult to pick up cues from their voice.'

4.3 Usage and age

The survey results indicated a number of differences in computer usage patterns across different age groups as is clear from Fig 5. The main age differences were between the very young, those in middle life, and those aged over 50.

For example younger people were more likely to own or use computers than older people, but where older people did have access to a computer they used them for longer periods and most particularly for work, a category which included household accounts and personal correspondence. The data also showed that, of those studied who used the Internet, those in the 31 to 50 age bracket used it more than the 16-30. This finding contrasts strongly with results from Kraut et al [8] who found that teenagers (particularly males) spent the highest number of hours per week using the Internet, compared to adults of both sexes.

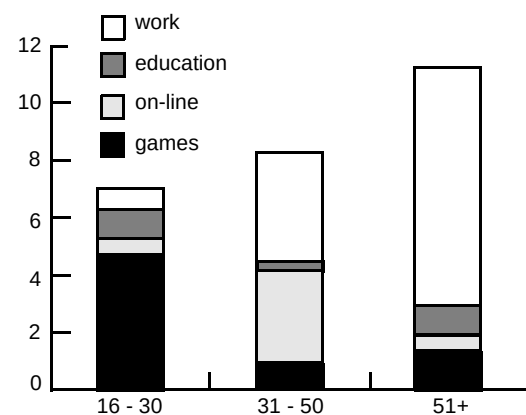


Fig 5 Mean PC usage for various reasons in hours per week by age.

Substantial age differences were also found for telephony. For example, 40% of people in their 20s and 30s had daily contact with relatives, compared to 30% of those in their 40s and 50s, and 21% of those aged over 60. A similar pattern of results was found in terms of frequency of daily telephone contact with friends. However, when call durations were examined the age distribution showed a different pattern. If the over 60s have the least frequent telephone contact with friends or relatives, overall they speak longest on the telephone as shown by Fig 6.

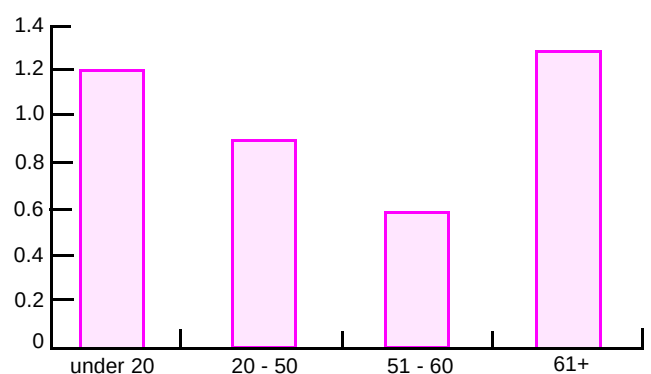


Fig 6 Mean hours per week on telephone by age.

4.4 Computers, television and the telephone — competition, substitution or multiplication?

Telephones, computers and televisions are not wholly competing technologies since computer games and TV and

Web browsing offer entertainment; telephones and e-mail are used for communication and TVs and the World Wide Web are used for information seeking.

The qualitative study indicated that telephone calls tended to be shorter when made from a room where the television was on. Analysis of the diary data found that television viewing behaviour was not strongly associated with computer usage. However, it is important to separate computer usage from Internet usage. For example Kraut et al [8] found that peak Internet usage coincided with prime-time television, which has implications for products offering Internet access over the television.

Not surprisingly, the data suggested that Internet usage appeared to boost total usage of the telephone system considerably, the majority of which was due to data calls. However, the studies also suggested that Internet users make more voice calls than non-Internet users and that high telephone contact with friends and high computer usage reinforce each other. Again it is currently unclear whether these differences are actually another manifestation of social class differences in ownership and usage of technologies rather than differences *per se*.

4.5 The impact of physical location on calling behaviour

The qualitative studies found that the location of the telephone modified call behaviour in several ways. Several participants mentioned privacy issues and commented that they were uncomfortable in making calls from a shared space such as the living room or kitchen where other household or family members were present or likely to make an appearance. One participant commented:

'I don't call my girlfriend from home very much because my mum is always there. Sometimes she [girlfriend] gets annoyed when she rings me and I won't discuss things but I can't because my mum's there. It's not that she's listening, it's just that it's uncomfortable.'

Comfort was another consideration, as calls were shorter when made from physically uncomfortable locations. However, in the case of shared households, comfort factors were often outweighed by a need for privacy and some participants reported taking the telephone to such unlikely locations as the bathroom and the stairs to preserve their privacy. The bedroom telephone was also used for this purpose even if it was not the caller's own room.

Figure 7 shows the breakdown of all calls made or received during the first wave of the home communications diary in terms of physical location.

The fact that nearly half (48%) of all calls recorded in diaries during the first week of the trial were made in the living room, and that 85% of households had a

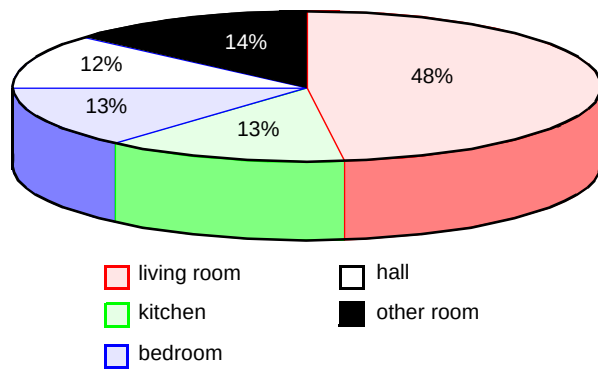


Fig 7 Location of calls made or received⁶.

communications product here, suggests that this room represents the centre of home communications. It is a shared family environment where individuals compete to use communications products and services.

Nearly three quarters of households (73%) had a communications product in a bedroom, the most popular being the corded telephone. This relatively high figure is reinforced by data from the survey, which indicated that where there is more than one telephone, the second is in the bedroom in at least 44% of households. Photographs from the diary study indicated that the most frequent location for communications products was next to the bed. However, only 13% of all calls recorded in the diaries occurred in the bedroom, which reinforces the finding from the qualitative study that the bedroom telephone only tended to be used when privacy was important.

Just over a half (51%) of households had a communications product in the kitchen, but only 13% of calls were made or received in this room. The photographs of communications products in the kitchen typically show wall-mounted corded telephones.

Over a third (38%) of households had a communications product in the hall, and 12% of calls took place here. The photographs typically show products located on small, low tables.

Over a third (38%) of households had an office. Typically they held a wealth of communications products and services (e.g. PC, fax, modem, telephony products) and were busy communications environments, although the living room still remained the communications 'hub'. The photographs provided useful evidence about how the fax machine is used within the home environment, e.g. it is not always found in the home office — rather they tend to be located next to the bed or in the hall.

The home communications diary study also collected call duration figures and it appeared that location made very

⁶ 'Other room' includes the home office, plus calls made from cordless phones.

little difference in terms of mean call durations, which ranged from 5.6 to 6 minutes, but the range of call lengths varied quite considerably.

4.6 *Behavioural adaptations to technology*

The qualitative study highlighted a range of social and behavioural adaptations and strategies employed by people to enable them to integrate communications technologies into their everyday lives. These strategies appear to centre on solutions to two key concerns and these are discussed in turn.

Privacy and access management

The telephone has gained a taken-for-granted, must-have position within the geography of the house migrating from an instrument placed most often in the hallway of the home, to the living room, kitchen, and even the bedroom. The home, however, is more than a physical space in which we eat, sleep and work, it is the sphere from which we conduct our relationships with family and friends and a place for 'privacy and independence' [9]. Different and varying rhythms are apparent within all our lives and this has an effect on when we declare ourselves to be available to those who would like to talk to us. As noted by de Sola Pool [3]: 'The telephone seems to have effects in diametrically opposed directions ... (it) invades our privacy with its ring but it protects our privacy by allowing us to transact affairs from the fastness of our homes.' Whether or not the telephone is viewed as a friend or foe depends on life circumstances and these are susceptible to changes that can happen very quickly on an hourly or daily basis.

For example, mothers at home with their children will be governed by the child's routine, which has an effect on the time of day that calls can be made; dropping off at school, mealtimes, and bath and bed routines are all 'no-go areas' for making, or indeed receiving, calls.

'I make most of my calls in the afternoon before the children come back, it's the only chance I get to have time for myself.'

Similarly, occupations can have an effect; those working shifts are differentially available at different times and such divergent and varying rhythms can change quite quickly.

The Home Communications diary study found that approximately half of all calls were made or received in the evenings between 5pm and 9pm, and within this slot 7pm was the most popular time for both incoming and outgoing calls. Within these broad figures, consciously or unconsciously people learn to adapt their calling behaviour to fit in with the routines they know exist within the lives of their family and friends. The qualitative study found that

different people declare themselves to be unavailable at different times of day, i.e. they are generally aware of when people they know well are likely to be busy and tend not to make calls at that time.

'I would never call my mum when EastEnders was on, or Gardeners' Question Time, she'd be really angry with me. I used to get it wrong all the time so now I usually call on Saturday mornings.'

If people are unfamiliar with the life rhythms of those they are calling, the telephone can become intrusive, and people devise their own systems and lines of defence against interruptions. The most common is the answering machine. Even though this device was designed to take calls when the recipient was out, participants describe using it as a screening device when they are at home.

'I've only started doing this recently but if the telephone rings and I'm eating, I get the kids to answer and tell whoever it is to call back. If it's my mum I'll speak to her but anyone else can call back.'

Not only do people declare themselves to be unavailable at different times of day, they may also be differentially available to different people. There was a need to signal short periods of time when people were unavailable so that callers could telephone back, or get through if the call was really urgent. The answering machine offers a partial solution, but many did not go to the trouble of changing their message to signal temporary unavailability. It was also felt answering machines indicated longer-term unavailability to the caller.

'I wouldn't put the answer machine on just because I was watching something on telly. The trouble is though that when the telephone does go in the middle of EastEnders I think it must be important because my friends know I'd be watching and they wouldn't interrupt me. So I think, oh, I'd better answer it and it turns out to be someone I don't know well enough to be rude to and say look, I'm watching EastEnders so I end up having to talk to them. It really annoys me.'

An alternative strategy is to employ a member of the household in the role of 'gatekeeper', defined as the person who is most likely to answer the telephone, and who acts as an intermediary or message taker between the caller and the call target. Gatekeeping occurs most frequently in family households and appears to be a fluid role most often undertaken by females. One family in the study had a well-defined gatekeeper (the mother) who always answered the telephone when she and her husband were alone in the house, but this role was taken over by their daughter when she came home from school. There was also some evidence of competition to claim the role of gatekeeper between teenage children.

In several cases the role of gatekeeper is assumed because other members of the household are indulging in call avoidance behaviour — hence the gatekeeper role often involves telling lies about the availability of the person to whom the caller wishes to speak.

‘He won’t answer the telephone unless he can possibly help it. Even when he knows it’s going to be about work I have to answer it and mouth who it is at him, and he’ll be making signals to say he doesn’t want to talk to them and I have to say he’s unavailable.’

Usage monitoring

Within many households at least one adult often took on the role of monitor, be it of the telephone or the computer.

In the case of the telephone, the call monitor decided when and how the telephone should be used and, not surprisingly, also tended to be the person responsible for paying the bill. Cost of calls was almost always the primary motivation in call monitoring. Instances of approaches to call monitoring included making family members underline all their calls on the telephone bill in different colours so that they would have a better idea of how much or how little they had used the telephone. Participants reported devising certain ‘house rules’ about telephone usage, which were not always popular with other household members and several call monitors said they suspected that rules were broken when they were absent.

‘When I come in from work I often catch one of them on the telephone, they always say it’s someone who’s rung them, or that it’s only local, or that they’ve only just rung, but the telephone bill usually says something different.’

However, a number of the participants admitted that where they played this role, they often did not extend the rules to their own behaviour. This resonates with the finding that people often made calls irrespective of time of day or potential cost rather than making a ‘rational’ cost decision. However, it also appeared that people were allocating a rough budget to their quarterly bill and adjusted their calling behaviour accordingly (i.e. not making calls at all) rather than making such decisions at the micro call-by call level.

4.7 A taxonomy of call types

Routine interaction is important to the maintenance of relationships [10], and becomes increasingly difficult to maintain over the years as local friends and family move away from each other [11]. Analysis of the qualitative data implied a new taxonomy of call types that provided a great deal of insight into consumer behaviour.

Duty calls

These calls are generally to family members and are made because the caller feels a sense of duty to keep in touch. They are often made on a weekly basis and tend to be made at regular times, usually during cheap rate periods and especially on Saturday mornings. In some cases such calls are scheduled in advance and any violation of this routine requires explanation by the person violating the arrangement. The most common forms of violation of this routine are not making the call at all, or calling outside of the pre-arranged time or day. Extra duty calls are also made on birthdays and anniversaries.

‘I always ring mum on Saturday mornings at about 10.30. We talk for about half an hour. I tell her what I’ve been doing and she tells me about her dogs and what the neighbours have been doing. I’m not really interested but she’d be upset if I told her that.’

Several female participants claimed that they were responsible for prompting their partners to make duty calls or in some cases even made them on their behalf. All participants said that the content of these calls concerned day-to-day events and that they are made because family ties carry a sense of obligation. Participants are also very accurate in predicting the duration of these calls (cross-checked with call records).

Maintenance calls

These calls are more frequent and more irregular than duty calls and are most likely to be made to friends rather than family. They are made at any time of day, and although such calls may be prompted by a specific reason, the real motivation is to maintain a friendship by keeping in contact. Maintenance calls made by men tend to be shorter and less frequent than those made by women and the content is different. Calls made by men usually have a focus or main topic of conversation and some men report delaying making such calls until they have something specific to talk about. Women, on the other hand, are happy to state explicitly that they are ‘just ringing for a chat.’ The comments from a male and female participant illustrate this difference:

Male participant: ‘There’s a couple of friends I ring just to keep in touch with but I usually do it when there’s something going on, something we can do together, I won’t just ring for a chat.’

Female participant: ‘I ring Anna [friend] two or three times a week, not for anything in particular, just for a chat.’

Pseudo-maintenance calls

Pseudo-maintenance calls are also made to friends in order to keep in contact and maintain the friendship. However, they are made deliberately when the caller knows that the call target is very unlikely to be there and the object of the call is simply to leave a message.

‘What I usually do is deliberately call Helen [friend] when I know she’s out and I can leave her a message on her answer machine. I say, hi, it’s me, just ringing to see how you are, then she knows that I’ve been thinking about her. That way I don’t actually have to talk to her right there and then when I’m busy, or I just don’t feel like a chat, but I have made contact.’

Pseudo maintenance calls were more common in women than in men and the reasons they gave for making such calls were also different. Male participants said that they did this because they did not have a bona fide reason for making the call, whereas female participants said that it was because they were too busy and didn’t have time for a chat, or were not in the mood for chatting. None of the participants reported this behaviour in calls to family members.

Grapevine calls

These calls are made in response to a piece of information or news received by the caller that they wish to pass on or find out more about. They are generally between family members but can also involve circles of friends. More females than males report this behaviour.

‘My mum rang me to tell me that my cousin had just had a baby, she was premature and there’d been all sorts of complications and things. I asked her about it but she’d got the news from her sister, via my aunt, and she didn’t really know what was happening so I telephoned my other cousin to find out what had really happened.’

Batch calls

This refers to a collection of calls being made at the same sitting. Such calling behaviour was often intentional on the part of the caller, especially when the calls had a single objective such as trying to make arrangements with a group of people, or pass on a piece of news to several others. Some participants also reported that they would stack together a group of unrelated calls that they needed to make and then make all the calls at a single sitting, this was most likely to occur in cheap rate periods. Many participants also reported that making a single call often prompted them to make several more, even though this had not been their intention when the first call was made. The

act of making a call seemed to act as a subconscious prompt.

‘When I picked up the telephone I was just going to make a quick call to my sister but when we’d finished talking I rang three other people. It’s as though once I’ve put the telephone down I think, who can I ring now?’

News calls

Participants reported that when they have to impart good news they are most likely to do this in cheap rate periods, and that this would often involve batch calls. Bad news on the other hand was spread to fewer people and calls were not influenced by the time of day. Participants also reported that good news calls tended to be shorter than bad news calls (possibly because sharing good news often involves batch calling), and that imparting bad news generated more incoming calls than good news.

‘When I split up with my girlfriend I didn’t really want to talk about it to anyone. I just rang my best mate and one of her friends who’d become my friend too.’

Multi-theme calls

Many calls were made with more than one purpose in mind. Unlike women, some male participants said that they waited until they had several things to relay to make the call ‘worthwhile’. Women reported that a single purpose was a more than adequate prompt to make a call and that calls then became multi-themed in the process of chatting.

‘I had to ring my sister to ask her to baby-sit and then I remembered that I’d left a teddy round at her place and I asked her if she’d bring that with her and we got on to talking about favourite toys we’d had when we were kids and what had happened to them.’

Shared calls

Several participants mentioned that they pass the telephone to another family member during the course of a call. This occurred most often when the call was from another member of the family, particularly in the case of grandparents talking to grandchildren and more rarely in three-way conversations with another member of the family ‘listening in’ and making comments on another telephone. These calls also take place where the telephone is a shared resource among friends who share common contacts and the caller may wish to speak to more than one person in the household. Such calls also occur opportunistically in shared households when the intended recipient of a call is not the person who answers the telephone but is known to the caller

who then engages them in conversation before the telephone is passed to the intended recipient.

'If the telephone rings and it's for me I'll often pass the telephone over to Jane [partner] so she can talk to them too, sometimes she ends up talking to them for longer than I do'.

Inter-generation calls

There were several examples of inter-generation calls and all of these were between family members. In one of the families, the husband spent some time away from home each week and rang his wife and two-year-old daughter every day. He saw this as vital to maintaining his relationships, particularly with his young child.

'Whenever I'm away I start the day with a call home, I love talking to Lucy [daughter] on the telephone even though she talks nonsense. I know she's not always listening to what I'm saying either because Rebecca [partner] told me that as soon as she hears my voice she starts cuddling the telephone and kissing it.'

Domino calls

Many participants said that if they make a call to a target who is unavailable they often make a call to a second, third or even fourth friend until they find someone to whom they can talk. This behaviour was much more common among female participants than males.

'Making a telephone call isn't like writing a letter where you just want to write to that person, if I feel like talking to someone and they're not there I'll just ring someone else, and if they're not in I'll try another friend. I suppose it's a bit like having a hierarchy of friends really'.

Information calls

These calls are short and to the point and are usually triggered by a need to make or check arrangements prior to a face-to-face meeting. They are made on a need-to-talk basis regardless of cheap rate periods and are very common in male calling patterns. Women report that they usually preface such calls with 'I'm just ringing to tell you...' to signal to the call target that they are not prepared to chat, and they will often promise to call back later for a chat even if they are going to see that person in the very near future.

'Most of my calls are about making arrangements, short and sweet really, see you at the pub at nine, that sort of thing.'

5. Summary

Even though the data described above is only the results of a one-year preliminary study, it has already built an extremely rich picture of how UK households are using, adapting and adapting to information and communications technologies.

The data supports the hypothesis that differences in communications patterns are as closely correlated with social class and income as are product ownership and, for example, health and life prospects. What is more, these differences seem to go deeper than simply access to and ownership of technologies but instead support Mayer's suggestion [12] that personal communication is an elastic need. Thus when people are able to afford to communicate more, either because their income increases or because the cost decreases, then they will do so.

The data has also highlighted age- and gender-related differences in communications behaviour and computer usage although it is unclear whether the latter are actually more to do with role and personality differences than gender *per se*.

The living room turned out to be the central communications hub with a high proportion of communications products being located here and being used frequently. The bedroom was a popular place to locate communications products although they were used less than those located in other rooms. Homes with an office make good use of communications products located in this room with office products being used to make calls rather than receive them.

Finally, the methods piloted during this preliminary phase have proven to be extremely successful. The communications diary gathered reported calling behaviour and insightful context of use even though it demanded a lot of effort to complete. As was illustrated in sections 4.1 and 4.2 the diary data also offered a way to cross-check the accuracy (or otherwise) of reported behaviour obtained via the questionnaire.

The photographs provided a rich and visual 'through the key hole' insight into the lives of the participating households. They show where communications products are located within the home environment, and in some cases how the life-styles of the people who use them are forced to fit around the technologies. For example, the photographs show that home offices suffer from 'space-famine' which means that communications products are located on top of speakers, under paper mountains, and on the office floor. In addition, products designed for the office (e.g. fax machines) were found in bedrooms and in the hall with a resulting trail of wires running around the house.

The qualitative research provided a rich picture of the communications lives of a small number of UK households and provided an invaluable way of accessing the reasons for the various communications patterns that emerged from the diary data and from the questionnaire. In addition, the studies described the wide range of behavioural adaptations not only in terms of communications behaviour but also of the wider social system which surrounds contemporary use of the telephone in UK households. With respect to commercial benefit, the studies highlighted a number of unmet needs of the participants, and the data suggested many potential service propositions, some of which have already been fed into marketing campaigns.

In summary, phase one of the Family Life in The Digital Home project has been a successful path-finding exercise. It has explored the ways in which our communications needs arise and are dealt with in the environments which we inhabit. ICT usage occurs within a physical space and within particular social situations within that space. In the domestic space, ICT usage has an impact upon other people who may be present at the time and the presence of other people can affect our usage. In shared households in particular, people make choices or have choices thrust upon them about how and where they use ICT. In building an understanding of ICT use in the home an insight can be gained into the ways in which our current needs are being met and of how future needs might be anticipated and fulfilled. The importance of the social context, i.e. locating ICT use in relation to people's roles and activities, and indeed, their own understanding of what is important in their lives, provides the kind of awareness that will enable the provision of innovative and appropriate services for the future.

6. Current and future research

Following the success of the pilot phase, the project is significantly expanding its scope to examine ICT usage across the spectrum of current and near-future technologies. In order to do this the project is currently setting up a panel of 1000 UK households (i.e. 2000 + individuals) which will be studied in detail for an initial period of 3 years but with the potential for indefinite extension into the future.

The panel is a nationally representative sample selected through a stratified random sample design. The panel will be replenished if participants withdraw but individuals who leave households will be followed wherever practically possible. The panel will over-recruit from the 'digital' population to artificially increase the sampling of households who already have PCs and modems in order to provide data on the growing usage of the Internet and other on-line services. The panel will be studied using a range of methods including those described earlier.

- Questionnaires

Questionnaires will be administered in the same manner as described in section 3.1 and will be repeated at twelve monthly-intervals.

- Time-use diaries

The time-use diaries piloted in phase one of the study will be given to the panelists for completion during the week following the questionnaire administration.

- Qualitative studies

Detailed interviews, shadowing and observational studies will be carried out on a small number of panellist households.

- On-line usage logging

A subsample of panellists will be instrumented with software that logs their PC and on-line usage.

- Customer data collection

The data obtained directly from panellists will be integrated with customer data held internally by BT.

These wide-ranging data sources will enable the project to not only build a more detailed picture of what UK households are doing with ICT, but also a unique view of how their usage changes over time as new products or services are melded into their daily lives.

It should be noted that even though the focus of the project's research is on ICT usage in the domestic context, this will also automatically provide a detailed picture of both teleworkers and home workers (those who run businesses from home), two of the most significant but least studied growth sectors for the ICT industry.

7. Conclusions

This paper has described results from the first stage of a long-term study of people's use of ICT products and services in the domestic environment. At one level the research has already generated significant knowledge that has led directly to commercial benefit. At a second it has validated a number of research methods intended to provide the data to build a rich picture of people's actual, in addition to their reported, behaviour. Finally, it has generated significant new scientific knowledge about the use of ICT products.

In the late twentieth century it is acknowledged that telecommunications needs are fundamental to the way we live our lives. Fast-moving technological, social and market changes in the last decade have meant that it is increasingly difficult to define customer needs in terms of standard services. A closer look now needs to be taken at the differences between people and of their adaptation of ICT to

fit increasingly complex lives. The project's next step is to build on these successes in order to develop a detailed understanding of how and why people purchase, adapt and adapt to information and communications technologies as UK society moves into the 21st century.

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